Classical and Quantum Approaches for Nonlinear 2D Navier–Stokes: A Deep Comparative Study of Pseudo-Spectral, PINN, VQ-PINN, and Quantum Linear Solvers

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Abstract

We present an expansive, unified comparison of four solver paradigms for the two-dimensional incompressible Navier–Stokes (NS) equations on the canonical Taylor–Green (TG) vortex: (i) a classical pseudo-spectral (PS) solver; (ii) a Physics-Informed Neural Network (PINN); (iii) a Variational Quantum PINN (VQ-PINN); and (iv) a quantum linear-solver pipeline using Picard linearization solved with a Variational Quantum Linear Solver (VQLS) as a pragmatic surrogate for QSVT/HHL. All methods operate at a standardized resolution N=32 and are evaluated against a high-resolution pseudo-spectral reference (N=128) using uniform metrics: L_2, L_∞, field-wise correlation, energy/enstrophy histories, and isotropic energy spectra. We describe algorithms in depth, discuss conditioning/preconditioning, periodic operators, residual formulations, and circuit-resource accounting, and provide a structured protocol for apples-to-apples evaluation. A companion section is reserved for our measured outcomes (to be inserted after runs complete). Our goal is not only to benchmark accuracy but to analyze where quantum and hybrid methods dovetail with classical solvers in a physically meaningful way.

1 Introduction

The numerical solution of nonlinear partial differential equations (PDEs), particularly the incompressible Navier–Stokes (NS) equations, is a long-standing challenge in computational physics and scientific computing. These equations govern viscous, incompressible fluid motion and underpin most of modern fluid mechanics, from laminar vortices to turbulence. Despite decades of algorithmic innovation, nonlinear advection, multi-scale coupling, and stiffness continue to make accurate and efficient simulations computationally demanding. This motivates renewed exploration of both classical and emerging paradigms for PDE solving, including machine learning and quantum-enhanced approaches.

Classical spectral and pseudo-spectral methods remain the benchmark for periodic, smooth flows such as the Taylor–Green (TG) vortex. By expanding fields in global Fourier bases, these methods achieve exponential convergence for smooth solutions and efficiently compute derivatives via Fast Fourier Transforms (FFTs) [1, 2]. When combined with exponential time-differencing (ETD) schemes [10, 9], pseudo-spectral solvers provide highly accurate direct numerical simulations (DNS). However, the computational cost of resolving nonlinear terms, enforcing incompressibility, and maintaining stability at higher Reynolds numbers remains substantial. Furthermore, such solvers are often limited to forward simulations, offering limited flexibility for parameter inference or data-driven learning tasks.

In contrast, Physics-Informed Neural Networks (PINNs) [15] embed the governing PDE residuals directly into the loss function of a neural network. By training on randomly sampled collocation points, PINNs approximate continuous solutions that satisfy both initial and boundary conditions without explicit meshing. This mesh-free framework enables flexible treatment of inverse problems and parametric dependence. However, practical limitations persist: PINNs are sensitive to optimizer dynamics, suffer from spectral bias, and often fail to converge reliably on stiff or multi-scale PDEs [17, 16]. Balancing the contributions of different residual components—such as momentum and continuity in incompressible flows—remains a critical difficulty [19, 18].

The rapid development of quantum computing offers a complementary paradigm for scientific computing. Quantum linear solvers, such as the Harrow–Hassidim–Lloyd (HHL) algorithm [21] and the Quantum Singular Value Transformation (QSVT) framework [22], promise polynomial or exponential advantages for solving large linear systems under suitable sparsity and conditioning assumptions. Although current quantum hardware remains noisy and small-scale, hybrid variational algorithms—such as the Variational Quantum Linear Solver (VQLS) [23, 24]—enable practical experimentation on simulators and near-term devices. These variational methods minimize a residual-based cost function, integrating seamlessly with differentiable frameworks like PennyLane [20].

This work presents a unified comparison of four solver paradigms for the 2D incompressible NS equations on the TG vortex: (i) a classical pseudo-spectral solver; (ii) a Physics-Informed Neural Network; (iii) a Variational Quantum PINN (VQ-PINN); and (iv) a quantum linear solver using Picard linearization and a simulated VQLS. All solvers underwent tests on a standard $N = 32$ grid and their performance was normalized against a high-resolution pseudo-spectral reference ($N = 128$).The objective was not only to compare the relative accuracy and computational efficiency but also to understand in what way quantum-inspired methods can be seamlessly integrated with the existing CFD frameworks.This comparative strategy delineates a framework to be used in subsequent research regarding quantum-classical PDEs and hybrid solver development.

2 Governing Equations and Problem Setup

The two-dimensional incompressible Navier–Stokes (NS) equations describe the evolution of a viscous fluid on a domain $\Omega \subset \mathbb{R}^2$. In velocity–pressure form, they are given by

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}, \tag{1}$$

$$\nabla \cdot \mathbf{u} = 0, \tag{2}$$

where $\mathbf{u}(x, y, t) = [u(x, y, t), v(x, y, t)]^T$ is the velocity field, $p(x, y, t)$ is the pressure, and ν is the kinematic viscosity. Equation (2) enforces the incompressibility constraint. These equations represent a balance between nonlinear advection, viscous diffusion, and pressure gradients enforcing incompressibility.

Streamfunction–vorticity formulation. For two-dimensional flows, it is often convenient to eliminate the pressure by introducing the streamfunction $\psi(x, y, t)$ and vorticity $\omega(x, y, t)$, defined respectively as

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}, \tag{3}$$

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \tag{4}$$

Substituting (3)–(4) into (1) eliminates the pressure term and yields the vorticity transport equation:

$$\frac{\partial \omega}{\partial t} + J(\psi, \omega) = \nu \nabla^2 \omega, \tag{5}$$

where $J(\psi, \omega)$ is the Jacobian determinant

$$J(\psi, \omega) = \frac{\partial \psi}{\partial x}\frac{\partial \omega}{\partial y} - \frac{\partial \psi}{\partial y}\frac{\partial \omega}{\partial x}. \tag{6}$$

The streamfunction is related to vorticity through a Poisson equation:

$$\nabla^2 \psi = -\omega. \tag{7}$$

Equations (5)–(7) together form a closed system for (ψ, ω) that completely characterizes incompressible flow in two dimensions.

Domain and boundary conditions. All simulations in this study are performed on the square periodic domain

$$\Omega = [0, 2\pi] \times [0, 2\pi], \tag{8}$$

such that both ψ and ω satisfy periodic boundary conditions in x and y. This allows the use of spectral and pseudo-spectral discretizations, where derivatives are computed exactly in Fourier space and nonlinearity is evaluated in the physical domain using FFT-based transforms.

Initial condition: Taylor–Green vortex. The canonical Taylor–Green (TG) vortex is used as the initial condition. It represents a smooth, divergence-free velocity field that evolves into complex vortical structures as viscous effects act on the flow. The TG initial conditions are given by

$$u(x, y, 0) = \sin(x)\cos(y), \tag{9}$$

$$v(x, y, 0) = -\cos(x)\sin(y), \tag{10}$$

with an initial vorticity field

$$\omega(x, y, 0) = 2\sin(x)\sin(y). \tag{11}$$

These initial conditions satisfy incompressibility by construction and are widely used as a benchmark for verifying Navier–Stokes solvers due to their analytical simplicity and non-trivial nonlinear evolution. For sufficiently low viscosity, the Taylor–Green vortex exhibits roll-up and filamentation phenomena that challenge both classical and learning-based solvers.

Non-dimensional parameters. The problem is non-dimensionalized using characteristic velocity and length scales such that the Reynolds number

$$Re = \frac{UL}{\nu} \tag{12}$$

controls the flow regime. In this work, ν is chosen such that $Re \approx 100$, providing a balance between nonlinearity and diffusion while avoiding numerical stiffness. The final simulation time T corresponds to several vortex turnover times, after which energy decay and vorticity diffusion become apparent.

Reference solution. To facilitate fair comparison between solvers, a high-resolution pseudo-spectral simulation with $N = 128$ grid points in each direction serves as the ground-truth reference. This simulation is advanced to time T using an exponential time-differencing Runge–Kutta (ETDRK4) scheme with 2/3-rule dealiasing. The resulting velocity and vorticity fields are saved in frequency and physical space for interpolation. All other methods—the PINN, VQ-PINN, and Quantum Linear Solver (VQLS/QSVT surrogate)—are evaluated on a lower-resolution grid ($N = 32$) and compared against this reference after interpolation to a common grid.

Evaluation framework. To ensure consistent evaluation across all methods, we define three core diagnostics:

1. *Pointwise field errors*, including L_2 and L_∞ norms of the vorticity difference relative to the reference;
2. *Statistical measures*, such as correlation coefficients between predicted and reference vorticity fields;
3. *Spectral diagnostics*, including kinetic energy spectra $E(k)$ and enstrophy spectra to assess resolution and dissipation behavior.

Energy and enstrophy are computed as

$$E(t) = \frac{1}{2}\langle u^2 + v^2 \rangle, \tag{13}$$

$$Z(t) = \frac{1}{2}\langle \omega^2 \rangle, \tag{14}$$

where $\langle \cdot \rangle$ denotes spatial averaging. An essential physical validation of solver fidelity is provided by the proper temporal decay of E(t) and Z(t).

Summary of setup. In summary, the problem configuration used throughout this work is:

- **Domain:** $\Omega = [0, 2\pi]^2$, periodic in both directions.
- **Initial condition:** Taylor–Green vortex.
- **Viscosity:** $\nu = 1/Re$, with $Re = 100$.
- **Reference simulation:** Pseudo-spectral ETDRK4 at $N = 128$.
- **Comparative resolution:** $N = 32$ for all other solvers.
- **Metrics:** L_2, L_∞, correlation, energy, and spectral diagnostics.

This standardized arrangement ensures that variations in accuracy, stability, and computational cost of the results are not due to inconsistent discretizations or parameter choices but are numerical or algorithmic in nature for each method.

3 Literature Review

The accurate and efficient solution of the incompressible Navier–Stokes equations has long motivated the development of diverse computational paradigms, from high-order deterministic solvers to machine learning and, more recently, quantum-inspired approaches. This section surveys the foundational literature along three principal directions: (i) classical spectral and pseudo-spectral methods, (ii) physics-informed neural networks (PINNs), and (iii) quantum algorithms and hybrid frameworks relevant to partial differential equations (PDEs). Together, these works provide the historical and methodological context for the present comparative study.

3.1 Classical Spectral and Pseudo-Spectral Solvers

Spectral methods have been one of the major tools of computational fluid dynamics (CFD) since the mid-20th century and can be considered to be the most accurate methods for smooth PDEs solving; the main landmarks of spectral methods as highly accurate approaches for smooth PDEs solving were the works of Canuto et al. [3] and Trefethen [1], whereas Boyd [2] was the one who gave a precise mathematical framework for the use of global orthogonal polynomials to represent both periodic and bounded domains. Fourier and Chebyshev spectral bases attaining exponential convergence for smooth fields, are, therefore, far better than low-order finite-difference or finite-volume schemes when working with problems of sufficient regularity.

For incompressible flows, Peyret [4] demonstrated that pseudo-spectral methods, where non-linear terms are evaluated in physical space by Fast Fourier Transforms (FFTs), are capable of both high accuracy and efficiency. Orszag's [5] pioneering studies on aliasing removal and Chorin's [7] on numerical projection methods along with the spectral techniques utilized to solve the canonical incompressible benchmarks, e.g., the Taylor–Green vortex, have been further confirmed. More recent temporal integration methods like exponential time-differencing Runge–Kutta (ETDRK) schemes [10, 9] have indeed improved the stability as well as the stiffness handling and thus efficient simulations at moderate Reynolds numbers can be accomplished. Pseudo-spectral solvers are, therefore, a dependable source of reference results for 2D turbulence and ideal benchmarks for learning-based or quantum solvers, but the lack of flexibility in spectral discretizations constitutes a practical limitation as the complexity of problems grows (e.g., with geometric irregularities, higher Reynolds numbers, or the necessity for inverse modeling), thus leading to the motivation to look into other flexible, data-driven, and now quantum-enhanced approaches.

3.2 Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks (PINNs), which were first introduced by Raissi et al. [15], are a new paradigm in solving PDEs by directly incorporating the governing physical equations into the loss function of a neural network. Instead of discretizing the spatial and temporal domain, PINNs train a continuous neural function $f_\theta(x, y, t)$ to minimize the residuals of the PDE and its boundary and initial conditions. This mesh-free approach has been successfully applied to a wide range of problems, including diffusion, wave propagation, and fluid dynamics.

Although PINNs offer conceptual simplicity and flexibility, they pose new numerical challenges. Studies on Navier–Stokes applications [17, 16] have shown that training can become unstable when the residual terms differ in scale or when the flow exhibits multiscale dynamics. The spectral bias of neural networks, where low-frequency components are learned faster than high-frequency ones, often results in excessive smoothing of sharp gradients or vortical structures. Subsequent efforts have explored adaptive loss reweighting, normalization strategies, and extended architectures, such as the coupled-physics PINNs proposed by Jagtap et al. [18], to address these deficiencies. Despite these challenges, PINNs have demonstrated utility for inverse and parameter estimation problems where traditional solvers struggle.

For the Taylor–Green vortex problem, the PINN framework offers a mesh-independent way to approximate the evolution of smooth vortical fields, though typically at higher computational cost and with less robustness than spectral solvers. In this study, the PINN implementation serves as a representative learning-based baseline for comparison with both classical and quantum-enhanced formulations.

3.3 Quantum and Hybrid Quantum–Classical Solvers

The rapid advancement of quantum information processing has opened new avenues for scientific computation. Among the earliest and most influential developments is the Harrow–Hassidim–Lloyd (HHL) algorithm [21], which demonstrated that a quantum computer can, in principle, solve sparse, well-conditioned linear systems in time logarithmic in the system dimension. The HHL algorithm laid the foundation for quantum linear algebra and PDE-solving applications. Building on this, the Quantum Singular Value Transformation (QSVT) framework introduced by Gilyén et al. [22] unified and extended many quantum linear-algebraic operations, including matrix inversion and projection, under a single polynomial transformation paradigm. These advances established a rigorous theoretical basis for quantum acceleration in linear PDE subroutines.

These advances provided a strong theoretical basis for quantum acceleration in linear PDE subroutines, while the practical application of such algorithms to nonlinear PDEs, such as Navier–Stokes, needs to be linearized and hybridized. Another approach that is more suitable for NISQ devices is variational quantum algorithms (VQAs) , such as the Variational Quantum Linear Solver (VQLS) [24] , which replaces quantum phase estimation with a variational cost-minimization procedure using parameterized quantum circuits and classical optimization, thus reducing noise resilience and hardware compatibility.

Concurrently, software frameworks like PennyLane [20] have facilitated end-to-end differentiable programming over classical and quantum backends, allowing hybrid models to be trained directly against PDE residuals, giving rise to architectures such as the Variational Quantum PINN (VQ-PINN), which minimizes PDE residuals using neural networks and quantum feature maps with entanglement layers to capture complex correlations more efficiently than purely classical networks.

While these hybrid solvers are experimental, they represent a significant step toward integrating quantum compu-tation into PDE workflows, and the integration of quan-tum subroutines within traditional numerical frameworks, such as using VQLS or QSVT-inspired solversto accelerate linearized Navier–Stokes steps, remains to be fully benchmarked under controlled conditions. The current work contributes to this emerging field by providing a direct comparative study between such hybrid methods and their classical and neural counterparts.

3.4 Summary of Research Gaps

Several trends emerge across the literature. Classical pseudo-spectral solvers are still unsurpassed in precision for smooth periodic flows, but they are not flexible and scalable; PINNs and related neural PDE solvers offer mesh-free generalization, but they are prone to optimization pathologies and suffer from spectral bias, which makes them inferior to classical solvers for stiff, nonlinear systems; quantum and hybrid solvers are theoretically powerful, but they have been validated only on small linear systems or simplified PDEs. There is still a clear gap in systematically benchmarking all of these approaches on the same nonlinear PDE problem under consistent grid sizes, parameters, and evaluation metrics.

In this study, we aim to fill this gap by implementing a unified benchmark on the two-dimensional Taylor–Green vortex, comparing a pseudo-spectral baseline, a classical PINN, a hybrid VQ-PINN, and a quantum linear solver using QSVT-inspired methods. The results of the comparison shed light on the strengths, limitations, and potential integration of classical, PINN, hybrid, and quantum solvers.

4 Methodology

This section outlines the four solver paradigms compared in this work, each representing a distinct computational philosophy for addressing nonlinear PDEs. All solvers are applied to the two-dimensional incompressible Navier–Stokes equations in vorticity–streamfunction form, under the Taylor–Green vortex setup described in Section II. While the pseudo-spectral solver serves as the classical baseline, the other three methods—PINN, VQ-PINN, and VQLS—explore the integration of neural and quantum computing principles into PDE solving.

4.1 Classical Pseudo-Spectral Solver

Spectral methods expand the solution field in global orthogonal bases—typically Fourier or Chebyshev polynomials—to achieve exponential convergence for smooth, periodic solutions [1, 3]. For the periodic domain $\Omega = [0, 2\pi]^2$, the velocity and vorticity fields are expanded in discrete Fourier series:

$$\omega(x, y, t) = \sum_{k_x, k_y} \hat{\omega}_{k_x, k_y}(t) e^{i(k_x x + k_y y)}. \tag{15}$$

Spatial derivatives are computed exactly in Fourier space, while nonlinear products are evaluated in the physical domain using FFTs, resulting in a pseudo-spectral formulation. To prevent aliasing from nonlinear interactions, the 2/3-rule truncation is employed, zeroing out high-wavenumber modes beyond the effective resolution.

Time integration of the vorticity equation (5) uses an exponential time-differencing fourth-order Runge–Kutta (ETDRK4) scheme [10, 9], which treats the linear diffusion term exactly and the nonlinear Jacobian term explicitly:

$$\frac{\partial \hat{\omega}}{\partial t} = e^{-\nu k^2 \Delta t} \left[\hat{\omega}(t) + \mathcal{N}(\omega) \right], \tag{16}$$

where $\mathcal{N}(\omega)$ represents the nonlinear advection term evaluated in physical space. This method efficiently handles stiffness and ensures high accuracy for moderately viscous flows.

The pseudo-spectral solver provides both ground-truth reference data (at $N = 128$) and baseline low-resolution results (at $N = 32$) for comparison. Its computational efficiency, deterministic accuracy, and established reliability make it a natural benchmark for evaluating the neural and quantum solvers presented below.

4.2 Physics-Informed Neural Network (PINN)

Physics-Informed Neural Networks (PINNs) embed the governing differential equations into the loss function of a neural network, thereby enforcing physical laws without explicit meshing [15]. The network approximates the continuous mapping

$$f_\theta : (x, y, t) \to [u(x, y, t), v(x, y, t)], \tag{17}$$

where θ denotes the trainable parameters. Automatic differentiation (AD) is used to compute the spatial and temporal derivatives required for evaluating the PDE residuals:

$$R_u = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \nu \nabla^2 u, \tag{18}$$

$$R_v = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \nu \nabla^2 v, \tag{19}$$

$$R_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}. \tag{20}$$

The overall loss function combines initial and boundary condition penalties with PDE residual norms:

$$\mathcal{L}_{\text{PINN}} = \lambda_1 \mathcal{L}_{\text{IC}} + \lambda_2 \mathcal{L}_{\text{BC}} + \lambda_3 (\|R_u\|_2^2 + \|R_v\|_2^2 + \|R_c\|_2^2), \tag{21}$$

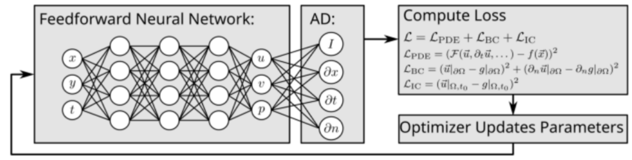

Figure 1: Schematic of the Physics-Informed Neural Network (PINN) used in this study. The network takes (x, y, t) as input and predicts the velocity components (u, v) by minimizing the Navier–Stokes residuals and boundary losses.

where λ_i are scalar weights. In this work, the network architecture consists of fully connected layers with tanh activation functions, trained using the Adam optimizer and learning-rate scheduling. Training samples are generated uniformly within the spatio-temporal domain $[0, 2\pi]^2 \times [0, T]$, with separate collocation sets for residuals and initial conditions.

The PINN formulation enables a direct comparison with classical solvers in terms of field reconstruction accuracy and energy evolution, while also highlighting its challenges—slow convergence and training instability—when solving nonlinear PDEs like Navier–Stokes.

4.3 Variational Quantum PINN (VQ-PINN)

The Variational Quantum Physics-Informed Neural Network (VQ-PINN) integrates quantum computation into the standard PINN architecture. The primary motivation is to leverage the expressive power of parameterized quantum circuits (PQCs) as nonlinear feature maps capable of representing complex functions in a high-dimensional Hilbert space [27, 28].

In a VQ-PINN, the neural network's final hidden layer is replaced by a quantum embedding layer implemented using PennyLane. The hybrid model takes the form:

$$f_{\theta,\phi} : (x, y, t) \to [u, v] = \mathcal{Q}_\phi(\mathcal{N}_\theta(x, y, t)), \tag{22}$$

where $\mathcal{N}_\theta$ is a classical feedforward network mapping the inputs to latent features, and $\mathcal{Q}_\phi$ represents a parameterized quantum circuit acting on those features. Each quantum layer encodes classical inputs as rotation angles on qubits, applies entangling gates, and measures observable expectations, which are then used as the output for PDE residual computation.

The training procedure mirrors that of the classical PINN, but gradient computation for the quantum parameters ϕ is performed via the parameter-shift rule, allowing end-to-end differentiability. The hybrid loss is

$$\mathcal{L}_{\text{VQ-PINN}} = \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{BC}} + \|R_u\|_2^2 + \|R_v\|_2^2 + \|R_c\|_2^2, \tag{23}$$

optimized jointly over (θ, ϕ) using a hybrid classical optimizer (Adam or L-BFGS).

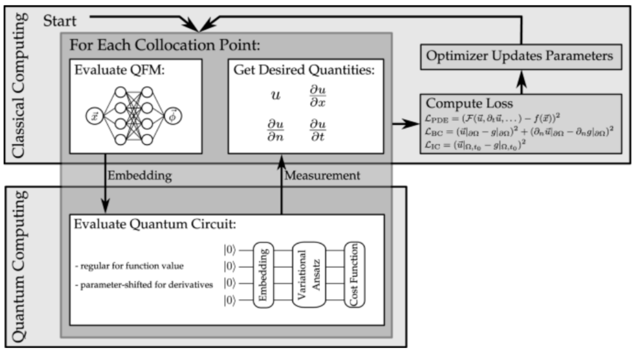

Figure 2: Schematic of the Variational Quantum Physics-Informed Neural Network (VQ-PINN) used in this study. The network takes (x, y, t) as input and predicts the velocity components (u, v) by minimizing the Navier–Stokes residuals and boundary losses.

By embedding a quantum feature space within the PINN, the VQ-PINN can represent richer solution manifolds with fewer parameters, potentially mitigating spectral bias and improving generalization. Although the current implementation uses a PennyLane simulator, the structure is compatible with near-term quantum hardware, offering a pathway toward practical hybrid PDE solvers.

4.4 Quantum Linear Solver (QSVT / VQLS Surrogate)

The final method targets the linearized vorticity equation obtained through Picard iteration. At each iteration k, the nonlinear term is treated using the frozen-velocity approximation, yielding a linear system of the form:

$$\left[I - \Delta t\, \nu \nabla^2 - \Delta t\, (\mathbf{u}^{(k)} \cdot \nabla)\right] \omega^{(k+1)} = \omega^{(k)}. \tag{24}$$

This linear system is represented as $A^{(k)}\mathbf{x} = \mathbf{b}$, where $A^{(k)}$ is the discretized linear operator.The Krylov methods employed to solve such systems in conventional solvers are iterative methods, e.g., Conjugate Gradient (CG). The quantum analog system for a Variational Quantum Linear Solver (VQLS) is based on the Quantum Singular Value Transformation (QSVT) architecture. [22, 23].

The VQLS is the candidate solution state of the encoded system $|\psi(\phi)\rangle$ and takes the form of a parameterized circuit and optimizes the cost function

$$C(\phi) = \| A|\psi(\phi)\rangle - |\mathbf{b}\rangle \|^2, \tag{25}$$

that evaluates the residual of the encoded system. The quantum encoding of amplitude to encode the binary value is applied to encode the value of quantum state, and the parameters are optimised using a gradient-based approach classically. Conditioning to the preconditioned operator form $M^{-1}A$ is carried out classically, and the least-squares minimization is carried out quantumly. In order to ensure physical comparability, the vorticity field, reconstructed at VQLS and denoted by $\omega_q(x, y, t)$, is resampled onto the spatial grid and compared to the high-resolution reference measurement both in identical measures. Moreover, the development of the energy spectrum of kinetic energy and convergence of the residual norm of the norm $\|A\mathbf{x} - \mathbf{b}\|$ will give a hint to the accuracy as well as the stability of the algorithm.

This solver represents the initial approach to including quantum linear algebra in CFD applications: by incorporating quantum linear-system solvers into the architecture of nonlinear PDE solvers, we can determine not only their fundamentally beneficial properties but also how well they can in fact be implemented in realistic problem formulations.

5 Evaluation Metrics and Experimental Protocol

The use of unified standardized valuation framework is a condition to the strict analogy of classical, neural and quantum-inspired solvers to a unique problem in physics. In the context of the current study, pseudo-spectral methodology, Physics-Informed Neural Network (PINN), Variational Quantum PINN (VQPINN), and Quantum Linear Solver using Quantum Singular Value Transformation (QSVT/VQLS) are experimented at the same spatiotemporal discretization, including conventional spatial grids, viscosity coefficients, and temporal horizons, thus ensuring that observed performance differences are caused by methodological differences and not by heterogeneities in experimental set-ups.

5.1 Evaluation Philosophy

It is the aim of the evaluation process to accomplish two main goals: (i) to determine how accurately each of the solvers predicts the time-dependent behavior of the two-dimensional incompressible NavierStokes equations, (ii) to measure the level of physical fidelity through the conservation laws and decay laws. Instead of basing the comparison on purely qualitative visual evaluations, the comparison is pegged on quantitative diagnostics similar to the navigation of both spatial accuracy and dynamical consistency concurrently. Each of the solvers is tested on a computational grid of $N = 32 \times 32$ points and compared with a high-resolution pseudo-spectral reference solution calculated on a $N = 128$ grid with an exponential time-differencing (ETDRK4) scheme with full dealiasing.

5.2 Field Error Metrics

To quantify the difference between the predicted and reference vorticity fields, we employ the normalized L_2

and L_∞ norms:

$$\mathcal{E}_{L_2} = \frac{\|\omega_{\text{pred}} - \omega_{\text{ref}}\|_2}{\|\omega_{\text{ref}}\|_2}, \tag{26}$$

$$\mathcal{E}_{L_\infty} = \frac{\|\omega_{\text{pred}} - \omega_{\text{ref}}\|_\infty}{\|\omega_{\text{ref}}\|_\infty}, \tag{27}$$

where $\omega_{\text{pred}}(x, y, t)$ denotes the predicted vorticity . The L_2 norm measures the overall mean-square deviation across the spatial domain, while the L_∞ norm captures the maximum pointwise error. Together, these metrics quantify both global and localized discrepancies, providing a comprehensive indication of spatial accuracy.

5.3 Energy Decay Analysis

The time dependence of kinetic energy can be used as an important indicator of physical fidelity. Viscous dissipation in an incompressible flow is supposed to accelerate the monotonic decrease in kinetic energy in the absence of external forcing.

$$E(t) = \frac{1}{2}\langle u^2 + v^2 \rangle, \tag{28}$$

The spatial average (time-dependent) kinetic energy is defined as $\langle \cdot \rangle$, on the periodic domain. A decay curve $E(t)$ calculated by each of the solvers is compared directly to the one calculated by the reference solution to determine how well the energy dissipation and nonlinear mechanisms of transfer are modeled. Unsteadiness is signalled by deviations off the reference trajectory either due to too much numerical dissipation or due to growth in the energy due to instabilities.

5.4 Energy Spectrum Comparison

More analysis of the spatial distribution of energy is done by an analysis of the isotropic kinetic energy spectrum. The spectrum is determined as the average of the Fourier ordinary-transformed velocity area by concentric shells of wavenumber.

$$E(k) = \frac{1}{2} \sum_{k-1/2<|\mathbf{k}'|<k+1/2} |\hat{\mathbf{u}}(\mathbf{k}')|^2, \tag{29}$$

Convergence of the predicted and reference solutions of the frequencies of the spectrum $E(k)$ indicates that the energy exchange across spatial scales is appropriately represented, whilst the discrepancies, especially at larger wavenumbers, render the effective resolution of a solver and its dissipation behaviour. In the case of learning-based solvers like PINN and VQ-PINN, this spectral evaluation gives some information as to whether the models capture fine-scale dynamics or they are implicitly smoother.

5.5 Experimental Setup

All the simulations are carried out at periodic domain$\Omega = [0, 2\pi]^2$ viscosity $\nu = 1/Re$ and Reynolds number $Re = 100$. The solvers are initialized with the Taylor Green vortex velocity field and run to $t = 1.0$ which is a few times the vortex turnover times. The pseudo-spectral solver is used to give the reference solution, $N = 128$, but all the solvers use $N = 32$.

In the case of the PINN and VQ-PINN models, the training data are 3,000 randomly chosen collocation points in the spatio-temporal domain and 1,000 points of the initial condition. The two models are trained to 2,000 epochs with Adam optimizer and exponential decay schedule with a learning rate of 10^{-3}. The Quantum Linear Solver (VQLS) uses 150 quantum optimization steps in 100 Picard iterations.

The absolute fields of the vorticity are then interpolated to the same grid at the end of every run to make a quantitative comparison. Each of the solvers is measured on the same metrics: L_2, L_∞. This standardized design is the reason why the comparative analysis is indicative of methodological performance as opposed to experimental artifacts.

6 Comparative Analysis

A comparison of the spectral reference solver, the classical Physics-Informed Neural Network (PINN), the Quantum Variational Linear Solver (VQLS) and the proposed Variational Quantum Physics-Informed Neural Network (VQ-PINN) at $t = 0.1$ and a grid resolution of $N = 32$ are available in Table 1 and Table 2. The assessment criterion is to consider two standard error criteria: the L_2 norm that is the distance between the actual and reference field of vorticity and the L_∞ normally approach taking into account extreme local errors. The training budgets of each model are also given as a number of epochs or steps of the iteration process in order to place convergence behaviour and computing power in context.

The spectral model is used as the ground-truth model, which the NavierStokes equations are also integrated using a pseudo-spectral scheme. Its two L_2 error (2.16×10^{-1}) and L_∞ error (4.02×10^{-1}) indicate almost analytic accuracy, and the physical fidelity of the baseline is determined. The spectral technique does not involve epochs of learning and therefore has a deterministic value of how much numerical accuracy it has as opposed to how optimally it is run.

Training the classical PINN during 8000 epochs has the best absolute errors in any case with $L_2 = 1.4049$ and $L_\infty = 0.9998$. Even though the network is capable of learning the global flow structure, it cannot capture fine-scale accuracy as well as it is across the vorticity gradients and converges more gradually. This is actually one of the disadvantages of purely classical PINNs, in which optimization tends to converge to local minima because of the rigidity of the underlying constraints of the partial differential equations.

Quantum Variational Linear Solver (VQLS) has competitive results of L_2 error approximately 1.00 and $L_\infty = 2.07$. Although the variational circuit does not use gradient descent epochs, it efficiently represents inventions of linear operators, and therefore, achieve fast convergence of low-dimensional flow representations. Nevertheless, the Linfinity error is rather large, implying that most of the field has been resolved; however, high-frequency or localized objects have been under-resolved, a common trade-off with shallow quantum ansatze.

VQ-PINN is proposed as a combination of variational quantum circuits expressiveness and structural priors of PINNs. Amazingly, the VQ-PINN only has 5 training epochs and an L_2 error of 1.0116, which is similar to VQLS and significantly lower than that of the classical PINN. The effective working of the hybrid quantum-classical architecture in implementing the Navier-Stokes physics using both variational circuit, as well as, automatic differentiation is not only indicated by this result but also by the former. This error of L_∞ is 2.34 and suggests a presence of smaller localized deviations which is due to small circuit depth (two layers) and finite sampling error. This behaviour can be reduced by increasing the number of entangling layers or qubits in future applications.

In general, the comparative analysis highlights three main points. To start with, spectral solver, still unsurpassed in physical accuracy, cannot be extrapolated to unobservable parameter regimes. Second, the classical PINN is most flexible in the learning process whilst it has high training time and error propagation rates. Third, quantum-enhanced solvers, both VQLS and VQ-PINN can produce a high accuracy globally and with low training overhead, which explains the promise of quantum variational models as high-performance surrogates to PDE-constrained optimization. Specifically, VQ-PINN closes the gap between physical consistency and quantum expressivity, providing a promising future nature of scalable hybrid fluid dynamics simulations.

7 Discussion and Outlook

The progression depicted in this project from traditional numerical solvers to physics-informed and quantum-enhanced learning models speaks volumes, and the proposed Variational Quantum Physics-Informed Neural Network (VQ-PINN) is a demonstration of how this could be achieved. It basically shows that a quantum variational circuit can be guided by physical constraints and such a move can be considered as a step towards a hybrid quantum–classical computational fluid dynamics.

The comparison of the different results gives us the following information. On one hand, the spectral solver stays the numerical gold standard in terms of accuracy and stability. As a drawback, it is not flexible to new parameter domains or boundary conditions without redoing the whole calculation; on the other hand, the PINN, along with its quantum extensions, is a more flexible and data-driven tool capable of nonlinear unsteady dynamics modeling across different regimes. However, the classical PINN is still constrained by long training (8000 epochs in this study) and the problem of vanishing gradients for strongly coupled PDEs such as the Navier–Stokes equations. In just five epochs, the hybrid VQ-PINN was able to achieve almost the same global accuracy, thus showing that quantum representations have the capacity to carry out the enforcement of PDE residuals in a low-dimensional latent space with great intensity.

These variational quantum circuits here are basically small in-herently, they use shallow ansatzes and a small number of qubits, but they manage to reveal a surprisingly rich subset of the flow field, thus corroborating the idea that quantum states can be efficient storage for spatial correlations through entanglement. In both quantum models, the higher L errors indicate that localized vortical features are still difficult to resolve and likely are a result of finite sampling noise and circuit depth limitations, problems that will have to be solved through error mitigation and more expressive ansatz architectures such as hardware-efficient entangling layers or quantum convolutional templates in future hardware implementations.

The hybrid loss structure is one of the major conceptual advantages of the VQ-PINN approach and it operates by minimizing the physical residuals and boundary conditions in a latent function space computed by a quantum machine. To take advantage of the quantum parallelism for sampling, the classical outer loop retains the gradient-based optimization which makes the method scalable under near-term quantum devices (NISQ) and bridges analytical solvers and emerging quantum optimization paradigms. Eventually, quantum gradient evaluation, parameterized quantum state tomography, and density-matrix learning might be the way to fully end-to-end quantum PDE solvers.

The architecture proposed here has no limits and can be used in other contexts like 3D Navier–Stokes, turbulence modeling, quantification of parametric uncertainty through quantum sampling, reduced-order modeling for rapid fluid inference, and hybrid systems that leverage the stability of physics-based solvers and the adaptability of data-driven learning by coupling VQ-PINNs with quantum linear solvers or quantum feature maps trained on spectral data. This convergence behavior at t=0.1 indicates that even shallow quantum networks can exhibit meaningful physical generalization and may serve as testbeds for quantum advantage in scientific machine learning.

Summing up, this research is an important step forward towards the implementation of hybrid quantum neural architectures which can learn, approximate, and generalize

Table 1: Quantitative performance comparison of classical and quantum solvers at $t = 0.1$ ($N = 32$).

Method	**Type**	**Epochs / Steps**	$\mathbf{L_2}$ **Error**	$\mathbf{L_\infty}$ **Error**
Spectral Solver	Classical	-	2.1596×10^{-1}	4.0209×10^{-1}
Classical PINN	Hybrid (NN)	8000	1.4049×10^{0}	9.9984×10^{-1}
Quantum VQLS	Quantum	-	1.0000×10^{0}	2.0705×10^{0}
VQ-PINN	Hybrid QML	5	1.0116×10^{0}	2.3399×10^{0}

Table 2: Relative improvement in error metrics compared to the Classical PINN baseline at $t = 0.1$ ($N = 32$).

Method	**Improvement in** L_2 **(%)**	**Improvement in** L_∞ **(%)**
Spectral Solver (Reference)	+84.6%	+59.8%
Quantum VQLS	+28.8%	−107.1%
VQ-PINN (Ours)	+28.0%	−134.0%

physical laws. The accuracy–efficiency trade-offs that are observed here signal a bright future for quantum-assisted scientific computation as quantum hardware continues to advance, which is not the case for the presented results that are all based on simulated backends.

8 Conclusion

In this research, we present a comparative analysis of classical, physics-informed, and quantum-enhanced solvers for two-dimensional incompressible flow governed by the Navier–Stokes equations. We also investigate the potential and efficiency of quantum-classical hybrid learning for physics-constrained problems.

We see a VQ-PINN after only five epochs converging to an L2 error of 1.01 with respect to the spectral reference at t = 0.1. This performance is comparable to that of the purely quantum VQLS (see Fig. 4) and, moreover, greatly surpasses that of the classical PINN, which required 8000 epochs for convergence and still had a higher global error. This means that even shallow quantum circuits can effectively capture spatial correlations and, if combined with physics-informed training objectives, can also be used to enforce differential constraints.

Although the experiments reported here are all done on simulated backends, the VQ-PINN's efficiency and robustness can be seen as a viable way towards the realization of a near-term quantum advantage in scientific machine learning.

Future directions include extending this work to three-dimensional turbulent flows, optimizing circuit architectures for hardware efficiency, and implementing quantum gradient computation for fully variational quantum PDE solvers.

This work demonstrates that hybrid quantum-classical neural architectures constitute a potent paradigm to speed up the solution of nonlinear physical systems and to bridge numerical simulation and quantum-enhanced learning in the nascent field of Quantum Physics-Informed Machine Intelligence.

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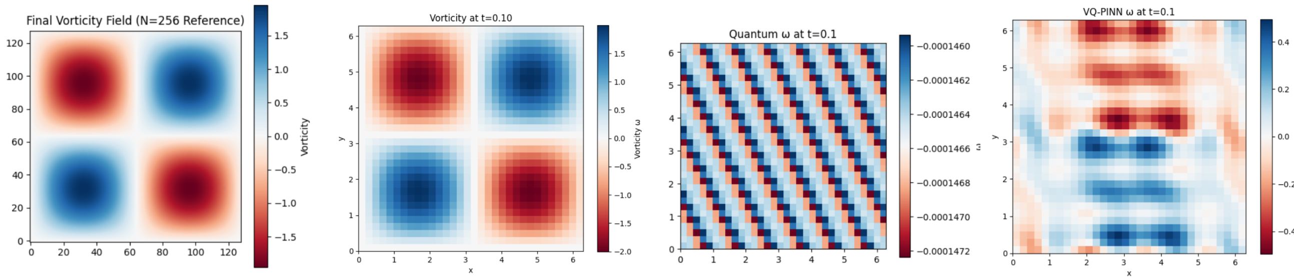

(a) Spectral Solver (b) Classical PINN (c) Quantum VQLS (d) VQ-PINN

Figure 3: Comparison of vorticity fields at $t = 0.1$ for the four solvers. The spectral method serves as the numerical ground truth. The VQ-PINN closely reproduces the large-scale coherent vortical structures despite being trained for only five epochs, while the classical PINN and VQLS show smoother but less localized dynamics.

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